

AMUTHEEZAN SIVAGNANAM

Visa Status: Lawful Permanent Resident / Green Card (USA), **Citizenship:** SRI LANKA

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SUMMARY

PhD Candidate and **Machine Learning Researcher** specializing in **Deep Reinforcement Learning** with **six** years of academic research and **two** years of industrial software development experience. Expertise in **scalable deep learning and reinforcement learning systems**, **ML training, and deployment**, complemented by a **strong publication record (h-index 6)**. Proficient in **Python, TensorFlow, PyTorch**, AWS, Agile and CI/CD practices, with hands-on experience in **optimizing ML models and pipelines**.

EDUCATION

Pennsylvania State University

Ph.D., Informatics — *Aug 2025*

Dissertation: *Application of Deep Reinforcement Learning to Solve Optimization Problems in Transportation Domains*

University of Houston

M.S., Computer Science — *Aug 2022*

University of Moratuwa

B.S., (Hons) Engineering (Computer Science and Engineering) — *Jan 2018*

Final Year Project: *Sentimental Analysis of Twitter using Semi-Supervised Approaches*

AREAS OF EXPERTISE

Reinforcement Learning, Optimization, Operational Research

SKILLS

Programming Languages: Python, C++, Java, C, JavaScript, PHP, MATLAB

Frameworks & Libraries: PyTorch, TensorFlow, Scikit-learn, Ray, RLLib, OpenAI Gym,

NumPy, pandas, matplotlib, CPLEX, Gurobi, Mosek, OR-Tools

Databases: SQL, MySQL, SQLite, MongoDB

Tools & Platforms: Amazon Web Services, Git, GitHub, Linux, Docker, Spark, CI/CD, Bash

Methodologies: Object-Oriented Development, Agile, Behaviour Driven Development

RESEARCH EXPERIENCE

Pennsylvania State University — Graduate Research Assistant

Applied Artificial Intelligence Lab

Aug. 2022 - May 2025

- Spearheaded **research projects** funded by the **U.S. Department of Energy (DOE)** and the **National Science Foundation (NSF)**, leading to advancements in **cost- and energy-efficient technologies**
- Studied **real-world decision-making problems** and identified **gaps** in existing solution approaches, leading to the development of **innovative strategies** for improved **decision-making**
- Developed **mathematical models** and formulated **problems** to address these challenges, resulting in more **accurate predictions** and solutions for **complex issues**
- Applied **artificial intelligence-based solutions** to tackle **real-world problems** and successfully **deployed** them in relevant **industries**
- Published **research findings** in **AI conferences (ICML)** and assisted in preparing **slides** for presenting results at **DOE and NSF meetings**
- Developed a **deep reinforcement learning (DRL)** method (using **DDPG**) for **proactive responder repositioning** in **emergency management**, achieving significantly **faster response times** while reducing **operational delays**
- Applied **DRL** (using **DQN**) to solve **online vehicle routing with advance booking**, enabling **real-time confirmations** within seconds

The University of Houston — Graduate Research Assistant

Resilient Networks and Systems Lab

Sep. 2019 - Aug. 2022

- Spearheaded **research projects** funded by the **U.S. Department of Energy (DOE)** and the **National Science Foundation (NSF)**, leading to advancements in **energy-efficient technologies**
- Studied **real-world decision-making problems** and identified **gaps** in existing solution approaches, leading to the development of **novel methodologies** for enhanced **decision-making**
- Developed **mathematical models** and formulated **problem statements** to effectively address these challenges, resulting in more **accurate predictions** and solutions for **complex issues**

- Applied **AI-based solution approaches** to **real-world problems** and successfully **deployed** them in relevant **industries**
- Published **research findings** in **AI conferences** (AAAI, IJCAI) and assisted preparing **slides** for presenting results at **DOE** and **NSF meetings**
- Introduced **heuristics** that reduced **annual energy costs** by **\$140K** for **public transit agencies** operating **mixed fleets of buses**
- Implemented a **deep reinforcement learning** (using **DQN**) approach that reduced **operational costs** by **20%** by enabling **online booking** for traditionally offline **Vehicle Routing Problems (VRPs)**
- Collected and analyzed **real-world data** using **Python** to identify **trends**, such as changes in **paratransit operations** before and after **COVID-19** and the impact of **vulnerability reward programs**

SOFTWARE ENGINEERING EXPERIENCE

LSEG Technology — Software Engineer

Post Trade Team

Jan 2018 – Jul 2019

- Developed application software using **Object-Oriented principles**, enhancing **code maintainability** and **scalability**
- Implemented **agile development practices**, such as **Scrum**, to improve **team collaboration** and **project delivery speed**
- Developed software solutions using **Java**, **Python**, and **C++**, improving **system performance** and **reliability**
- Introduced **Unit Testing** for **Libraries** in **Post Trade C++ Code**
- Modified **database structures** and tested using **BDD-based testing**, ensuring **data integrity** and **system functionality**
- Identified and replaced **duplicated error codes** with valid new error codes, improving **system reliability**
- Developed a **dynamic scripting object (DSO)** for **front-end end-to-end testing** using **Java**, enhancing **test coverage** and **efficiency**
- Analyzed and modified **back-end regression scripts** to work with both old and new **testing frameworks**, ensuring **compatibility** and reducing **testing time**
- Developed a new **report generation system** for the testing framework, **automating email dispatch** at the end of regression and improving **communication efficiency**
- Completed **code integration tasks** related to **back-end regression**, streamlining the **development process** and enhancing **system performance**
- Updated **automatic updates** to **auto-generated codes** based on **Database changes** using **Integration Plans**
- Contributed to **code integration** and **deployment plans**, ensuring **seamless software updates** and minimizing **downtime**
- Participated in **professional training programs** conducted by **Millennium IT Software** and **Post Trade Team**

- Worked on **Front-End Development** for both **Product** and **Solution** which consists of **Enhancement, Bug Fixing, Merging** and **Introducing new features**

WSO2 Lanka PVT Ltd — Software Engineering Intern

Data Analytics Team

Jul 2016 – Dec 2016

- Developed an HL7 Monitoring Solution by integrating **WSO2 ESB, DAS, and BAM**, enabling real-time and batch processing of healthcare data
- Developed an alert generation system analyzing descriptive **HL7/FHIR** data to monitor **disease outbreaks**, triggering timely email and SMS notifications
- Created **Spark scripts** for batch analytics and **Siddhi execution plans** for real-time alerts, including disease outbreak and patient wait-time notifications
- Engineered a mechanism to evaluate **hospital functionality** by assessing **admission and discharge** messages, including bed and oxygen cylinder availability
- Implemented interactive dashboards using **Jaggery, JavaScript, jQuery, Leaflet.js, and DataTables**, featuring gadgets like charts, maps, and tables
- Utilized **HAPI library and test panel** to simulate HL7 v2 messages and validate the monitoring pipeline end-to-end
- Packaged the entire monitoring solution as a **Carbon Application (CApp)**, bundling event streams, receivers, publishers, and visualization artifacts
- Attended workshops and gained hands-on experience with **Git, MSF4J (Microservices for Java)**, and **WSO2 product architecture** for enterprise middleware solutions

PUBLICATIONS

CONFERENCE PROCEEDINGS

[C7] **Sivagnanam, A.**, Pettet, A., Lee, H., Mukhopadhyay, A., Dubey, A., & Laszka, A. (2024). Multi-agent reinforcement learning with hierarchical coordination for emergency responder stationing. In *Proceedings of the International Conference on Machine Learning (ICML-24)*.

[C6] R. Sen, **A. Sivagnanam**, A. Laszka, A. Mukhopadhyay and A. Dubey, "Grid-Aware Charging and Operational Optimization for Mixed-Fleet Public Transit," *2024 IEEE 27th International Conference on Intelligent Transportation Systems (ITSC)*, Edmonton, AB, Canada, 2024, pp. 4172-4179.

[C5] Pavia, S., Rogers, D., **Sivagnanam, A.**, Wilbur, M., Edirimanna, D., Kim, Y., Mukhopadhyay, A., Pugliese, P., Samaranayake, S., Laszka, A., & Dubey, A. (2024). SmartTransit.AI: A dynamic paratransit and microtransit application. In *Proceedings of the Thirty-Third International Joint Conference on Artificial Intelligence (IJCAI-24)* (pp. 8767–8770).

[C4] Atefi, S., **Sivagnanam, A.**, Ayman, A., Grossklags, J., & Laszka, A. (2023, May). The benefits of vulnerability discovery and bug bounty programs: Case studies of Chromium and Firefox. In *Proceedings of the ACM Web Conference* (pp. 2209–2219).

[C3] **Sivagnanam, A.**, Kadir, S. U., Mukhopadhyay, A., Pugliese, P., Dubey, A., Samaranayake, S., & Laszka, A. (2022, July). Offline vehicle routing problem with online bookings: A novel problem formulation with applications to paratransit. In *Proceedings of the Thirty-First International Joint Conference on Artificial Intelligence (IJCAI-22)* (pp. 3933–3939).

[C2] **Sivagnanam, A.**, Ayman, A., Wilbur, M., Pugliese, P., Dubey, A., & Laszka, A. (2021, May). Minimizing energy use of mixed-fleet public transit for fixed-route service. In *Proceedings of the AAAI Conference on Artificial Intelligence (Vol. 35, No. 17, pp. 14930–14938)*.

[C1] Ayman, A., Wilbur, M., **Sivagnanam, A.**, Pugliese, P., Dubey, A., & Laszka, A. (2020, September). *Data-driven prediction of route-level energy use for mixed-vehicle transit fleets*. In *Proceedings of the 6th IEEE International Conference on Smart Computing (SMARTCOMP)* (pp. 1–8). IEEE.

JOURNALS

[J2] Wilbur, M., Ayman, A., **Sivagnanam, A.**, Ouyang, A., Poon, V., Kabir, R., Vadali, A., Pugliese, P., Freudberg, D., Laszka, A., & Dubey, A. (2023). Impact of COVID-19 on Public Transit Accessibility and Ridership. *Transportation Research Record*, 2677(4), 531-546.

[J1] Ayman, A., **Sivagnanam, A.**, Wilbur, M., Pugliese, P., Dubey, A., & Laszka, A. (2022). Data-driven prediction and optimization of energy use for transit fleets of electric and ICE vehicles. *ACM Transactions on Internet Technology*, 22(1), Article 7, 1–29.

WORKSHOPS

[W1] **Sivagnanam, A.**, Atefi, S., Ayman, A., Grossklags, J., & Laszka, A. (2021). On the benefits of bug bounty programs: A study of Chromium vulnerabilities. In *Workshop on the Economics of Information Security (WEIS)* (Vol. 10).

BOOK CHAPTERS

[BC1] Wilbur, M., **Sivagnanam, A.**, Ayman, A., Samaranayake, S., Dubey, A., & Laszka, A. (2023, August). Artificial intelligence for smart transportation. In **Y. Vorobeychik & A. Mukhopadhyay (Eds.)**, *Artificial Intelligence and Society* (book chapter). ACM Press.

PROJECTS

PHD RESEARCH PROJECTS

Emergency Responder Stationing

November 2022 – June 2024

- Developed a novel hybrid approach combining **Deep Reinforcement Learning (DDPG)** and **Combinatorial Optimization** for real-time emergency response decision-making
- Designed a **hierarchical control framework** with DDPG agents for **city-scale redistribution** and **region-scale reallocation** of emergency responders
- Built a **Transformer-based actor network** to handle variable numbers of responders and depots in region-level tasks
- Ensured **feasible and exact mapping** from continuous to discrete actions using **min-cost flow** and **max-weight matching**, preserving gradient flow during training.
- Integrated **low-level critics** to provide feedback on high-level coordination actions, enhancing stability and performance
- Achieved **1000x faster decision-making** and reduced response delays by **5–13 seconds** on real-world datasets

The Benefits of Vulnerability Discovery and Bug Bounty Programs

February 2020 – May 2023

- Conducted a comprehensive empirical analysis of **Chromium Vulnerability Reward Program (VRP)** reports, focusing on trends across stable and development versions
- Provided evidence that **vulnerabilities in stable releases are harder to detect**, validating the **security impact** of structured VRPs
- Demonstrated that **bug-bounty programs complement internal security teams** by uncovering a **diverse range of vulnerability types** beyond in-house detection capabilities
- Delivered **actionable insights** to enhance bug-bounty effectiveness, including **targeted guidance for bug hunters** toward vulnerabilities with higher real-world exploitation potential

Offline Vehicle Routing Problem with Online Bookings

February 2020 – July 2022

- **Introduced a novel problem formulation:** the **Offline Vehicle Routing Problem with Online Bookings**, which blends the **scalability of Offline VRP** with the **real-time responsiveness of Dynamic VRP**
- Modeled real-world paratransit operations where **riders book trips in advance** but expect **tight pickup time windows at booking time**, capturing a unique **offline-online hybrid decision-making** setting
- Developed a **Deep Reinforcement Learning-based policy** that learns to assign optimal time windows under demand uncertainty and booking-time constraints.
- Integrated an **anytime VRP solver** to incrementally **refine and improve route plans between bookings**, enabling better long-term efficiency
- Achieved **20–40% cost reduction** over baseline methods with naive window assignments, as demonstrated through extensive **experiments on real-world datasets**

Minimizing Energy Use of Mixed-Fleet Public Transit for Fixed-Route Service

August 2019 – May 2021

- **Formulated a mathematical model** to **optimize energy consumption** for public transit agencies operating **mixed fleets of Electric Vehicles (EVs) and Internal Combustion Engine Vehicles (ICEVs)**
- Transformed the model into an **Integer Programming (IP) formulation** to obtain **exact optimal solutions** for small to medium-sized problem instances
- Developed a scalable **solution approach combining heuristics and meta-heuristics** (e.g., local search, genetic algorithms) to solve **large-scale instances efficiently in polynomial time**

UNDERGRAD RESEARCH PROJECT

Sentimental Analysis of Twitter using Semi-Supervised Approaches

November 2016 – November 2017

- Developed a **sentiment analysis model** to classify tweets into **positive, negative, and neutral** categories
- Addressed **limited and noisy labeled data** by applying **semi-supervised learning techniques**, including **self-training, co-training**, and **lightweight ensemble methods**

- Initiated training with a **small hand-labeled dataset**, and iteratively expanded it by **automatically labeling high-confidence tweets**
- Refined the classifier through **multiple training iterations**, improving accuracy and robustness on real-world social media data

HL7 Monitoring System

July 2016 – December 2016

- **Built an end-to-end HL7/FHIR monitoring system** using WSO2 ESB, DAS, and BAM for both real-time (Siddhi) and batch (Spark) analytics of healthcare data
- **Designed an alert system** to detect disease outbreaks and long patient wait times, delivering **email and SMS notifications** based on HL7/FHIR data streams
- **Engineered hospital functionality assessments** by analyzing admission/discharge events, tracking resources like **bed and oxygen cylinder availability**
- **Developed interactive dashboards** with **Jaggery, JavaScript, Leaflet.js**, and **DataTables**, and packaged the solution as a **WSO2 Carbon Application (CApp)** for easy deployment

TALKS

IJCAI (Vienna, Austria)	2022
For the paper: Offline Vehicle Routing Problem with Online Bookings: A Novel Problem Formulation with Applications to Paratransit	
WEIS (Virtual Workshop)	2021
For the workshop paper: On the Benefits of Bug Bounty Programs: A Study of Chromium Vulnerabilities	
AAAI (Virtual Conference)	2021
For the paper: Minimizing Energy Use of Mixed-Fleet Public Transit for Fixed-Route Service	

SERVICES

Program Committee Member

International Joint Conference on Artificial Intelligence (AI For Social Good)	2025
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Reviewer

International Joint Conference on Artificial Intelligence (AI For Social Good)	2024
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Reviewer

AI4Research Workshop @IJCAI2024

2024

Auxiliary Reviewer

22nd International Conference on Autonomous Agents and Multiagent Systems

2023

AWARDS AND HONORS**Information Security Quiz 2015 - Runners Up**

2015

Issued by Sri Lanka Computer Emergency Readiness Team(SLCERT)
and Information and Communication Technology Agency of Sri Lanka(ICTA)

Mahapola Scholarship

2014 - 2017

REFERENCES**Dr. Aron Laszka (PhD Supervisor)**Position: **Assistant Professor**Affiliation: **Pennsylvania State University**Email: aql5923@psu.eduPhone: **814-865-1551**Website: <https://aronlaszka.com>**Dr. Abhishek Dubey**Position: **Associate Professor**Affiliation: **Vanderbilt University**Email: abhishek.dubey@vanderbilt.eduPhone: **615-322-8775**Website: <https://abhishekdubey.bio>**Dr. Weidong Shi**Position: **Associate Professor**Affiliation: **University of Houston**Phone: **713-743-3045**Email: wshi3@uh.edu